Importance sampling (Reweighting)

Reject, Rewight, Repeat "

> Goals: Find the integral

1) $A = \int a(\theta) f(\theta) d\theta / \int f(\theta) d\theta$

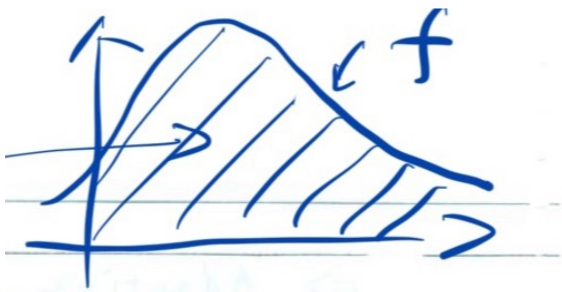
$= E(a(\theta))$, where $\theta \sim \frac{f(\theta)}{\int f(\theta) d\theta}$

where, $f(\theta)$ is unnormalized

$f(\theta) = \pi(\theta) P(x|\theta)$

C_f

2) $C_f = \int f(\theta) d\theta$



We can't draw $\theta \sim f(x)$

instead, we can draw $\theta \sim g(\theta)$

where, $g(\theta)$ is another unnormalized

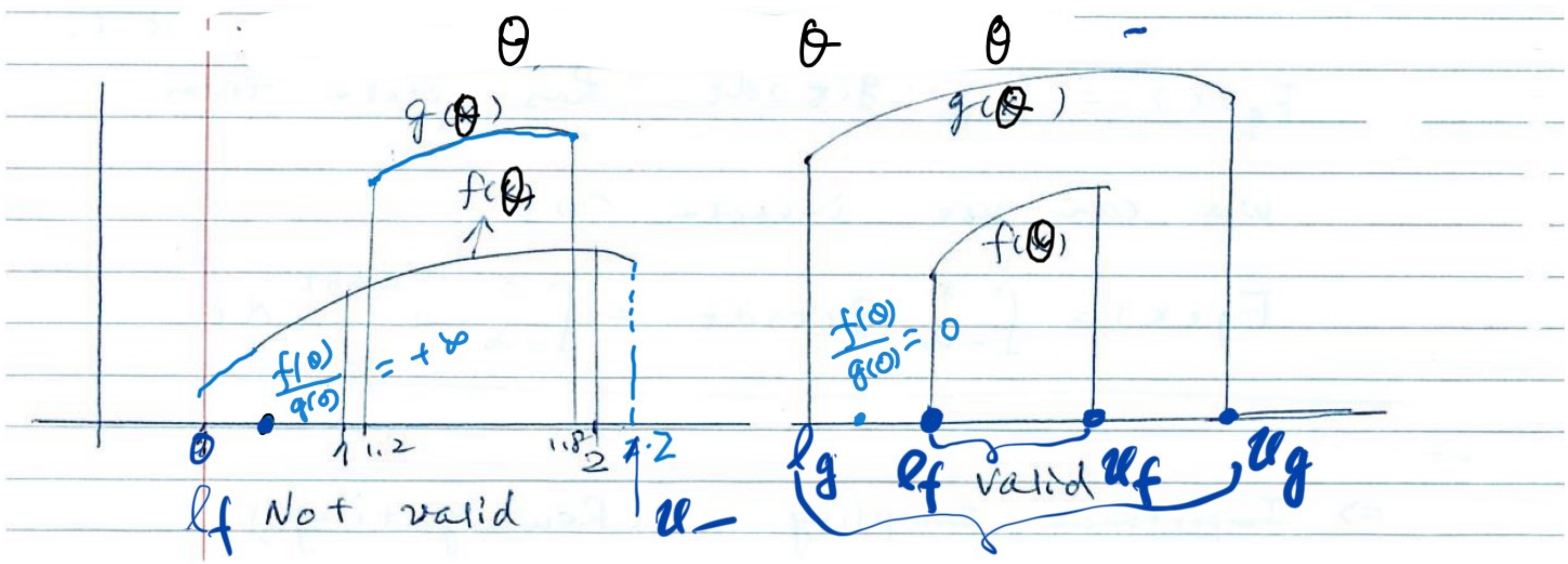
function, we hope $\frac{f(\theta)}{g(\theta)} \approx a$ (constant)

Requirement for $g(\theta)$:

$\{\theta \mid g(\theta) > 0\} \supseteq \{\theta \mid f(\theta) > 0\}$

Example of Supports of g and f

$(l_g, u_g) \geq (l_f, u_f)$ $\quad\quad$ $(l_g, u_g) \geq (l_f, u_f)$



Importance Reweighting Formula

1) draw $\theta_1, \cdots \theta_n \sim g$

2) $\widehat{\frac{C_f}{C_g}} : \sum_{i=1}^{n} \frac{f(\theta_i)}{g(\theta_i)} / n$, where $W(\theta) = \frac{f(\theta)}{g(\theta)}$

$$= \frac{\sum_{i=1}^{n} w(\theta_i)}{n}$$

$$\hat{A} = \frac{\sum_{i=1}^{n} a(\theta_i) \frac{f(\theta_i)}{g(\theta_i)}}{\sum_{i=1}^{n} W(\theta_i)} = \frac{\left[\sum_{i=1}^{n} a(\theta_i) w(\theta_i)\right] / n}{\left[\sum_{i=1}^{n} w(\theta_i)\right] / n}$$

where, $w(\theta_i) = \frac{f(\theta_i)}{g(\theta_i)}$

$$\hat{A} = \sum_{i=1}^{n} a(\theta_i) \cdot \tilde{W}(\theta_i)$$

$$\tilde{W}(\theta_i) = \frac{W(\theta_i)}{\sum_{i=1}^{n} W(\theta_i)}$$

$$\sum_{i=1}^{n} \tilde{W}(\theta_i) = 1$$

e.g.

	θ_1	θ_2	θ_3
$W(\theta)$	100	40	60
$\tilde{W}(\theta)$	0.5	0.2	0.3

Derivation of Importance Reweighting Formula

$$E_f(a(\theta)) = \int_{l_f}^{u_f} a(\theta)\frac{f(\theta)}{C_f}\,d\theta = \frac{1}{C_f}\int_{l_f}^{u_f} a(\theta) f(\theta)\,d\theta$$

$$= \frac{C_g}{C_f}\int_{l_f}^{u_f}\left(a(\theta)\frac{f(\theta)}{g(\theta)}\right)\cdot\frac{g(\theta)}{C_g}\,d\theta = \frac{C_g}{C_f}\cdot\int_{l_g}^{u_g} a(\theta)\cdot\frac{f(\theta)}{g(\theta)}\frac{g(\theta)}{C_g}\,d\theta$$

requiring $\{\theta \mid g(\theta) > 0\} \supseteq \{\theta \mid f(\theta) > 0\}$

$$= E_g\left(a(\theta)\cdot\frac{f(\theta)}{g(\theta)}\right) \Big/ \left(\frac{C_f}{C_g}\right)$$

$$= E_g\left(a(\theta)\cdot\frac{f(\theta)}{g(\theta)}\right) \Big/ \frac{C_f}{C_g}$$

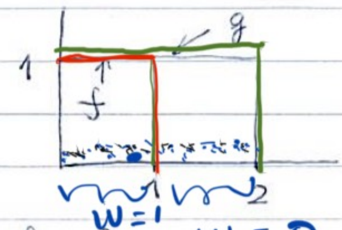
$$W(\theta) = \frac{f(\theta)}{g(\theta)} \quad \text{is called importance weight}$$

$$\frac{C_f}{C_g} = \int \frac{f(\theta)}{g(\theta)}\cdot\frac{g(\theta)}{C_g}\,d\theta = \frac{\int f(\theta)\,d\theta}{\int g(\theta)\,d\theta}$$

$$= E_g\left(\frac{f(\theta)}{g(\theta)}\right)$$

example 1) $f(\theta) = I(0 \le \theta \le 1)$

$g(\theta) = I(0 \le \theta \le 2)$



$C_f = 1$

$C_g = 2$

support of $g \geq$ support of f

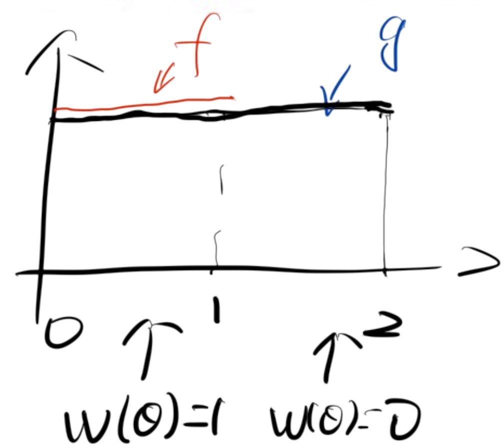
We want to estimate $\frac{C_f}{C_g}$ (true $= \frac{1}{2}$)

suppose $\theta_s \sim g\,[\text{unif}([0, 2]))]$

$$W(\theta_s) = \frac{f(\theta_s)}{g(\theta_s)} = \begin{cases} 1 & \text{if } 0 \le \theta_s \le 1 \\ 0 & \text{if } 1 < \theta_s \le 2 \end{cases}$$

$$\widehat{\left(\frac{C_f}{C_g}\right)} = \sum_{s=1}^{S} W(\theta_s) \Big/ S \approx \frac{1}{2}$$

$$\hat{A} = \frac{\sum_s a(\theta_s) W(\theta_s)}{\sum_s W(\theta_s)}$$



$$= \frac{\sum_{0<\theta_s<1} a(\theta_s)}{\sum_s I(0<\theta_s<1)}$$

$$= \frac{1}{n_E} \sum_{0<\theta_s<1} a(\theta_s), \text{ where } n_E = \#(0<\theta_s<1) \approx \frac{1}{2} S$$

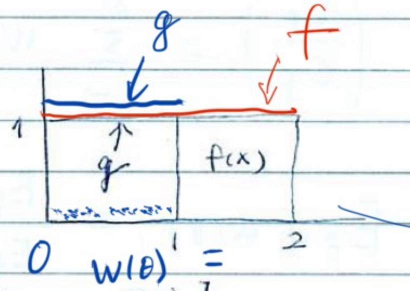
example 2) $f(\theta) = I(0 \le \theta \le 2)$

$g(\theta) = I(0 \le \theta \le 1)$

support of $g \subset$ support of f

We want to estimate

$\dfrac{C_f}{C_g}$ (true ≤ 2)

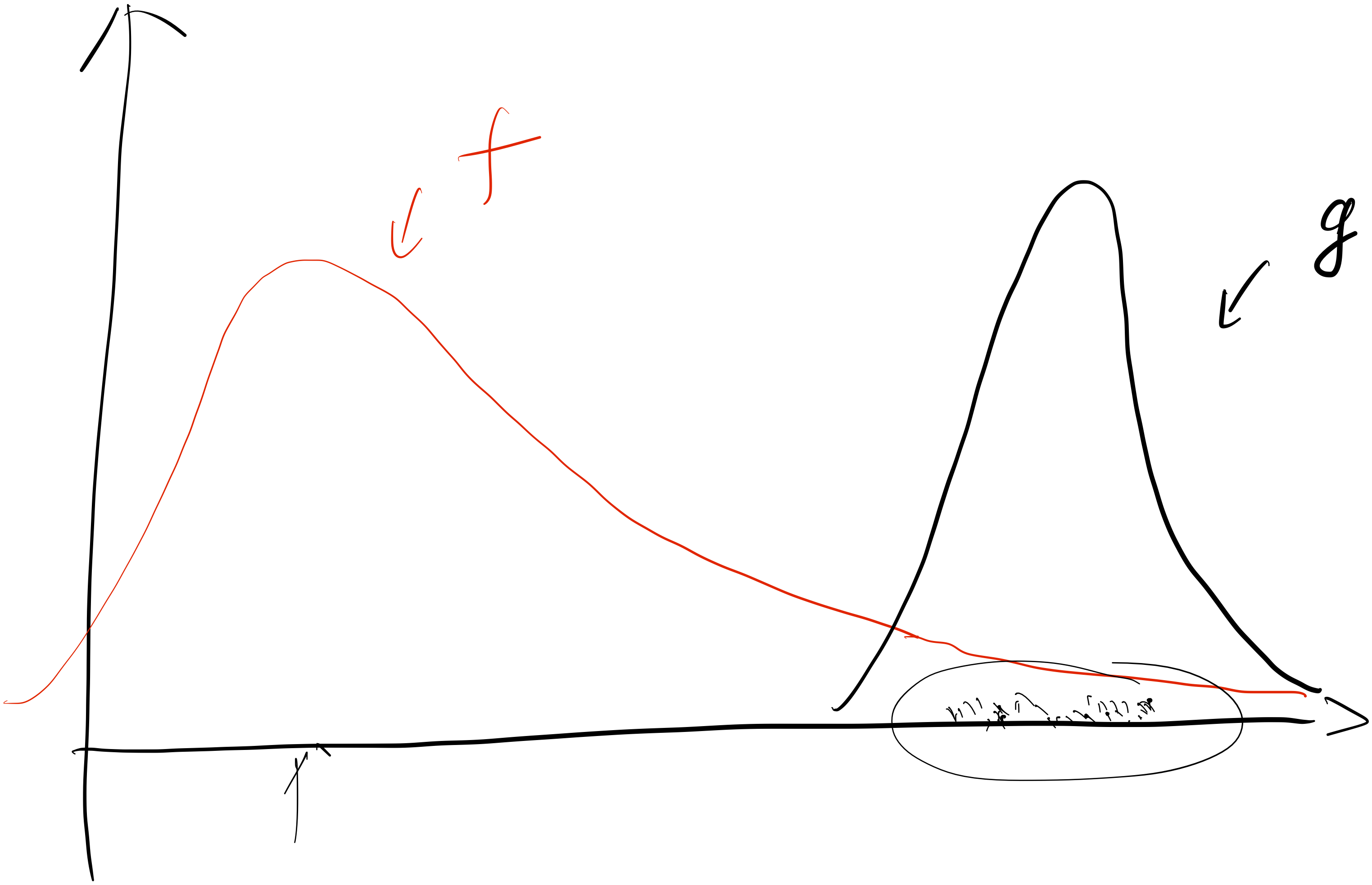


suppose $\theta_s \sim g\,[\,\text{unif}([0, 1])\,]$

$$W(\theta_s) = \frac{f(\theta_s)}{g(\theta_s)} = 1$$

$$\widehat{\left(\frac{C_f}{C_g}\right)} = \frac{\sum_{s=1}^{S} W(\theta_s)}{S} = 1 \ne 2 \quad (\text{true value})$$

Example of Poor Importance Sampling



Effective sample size :

$$S_{eff} = \frac{1}{\sum_{s=1}^{S} \tilde{W}^2(\theta_s)}$$

where, $\tilde{W}(\theta_s) = \dfrac{W(\theta_s)}{\sum_{s=1}^{S} W(\theta_s)}$

(Not reliable)

A special case $\quad \tilde{W}(\theta_s) = \dfrac{C}{S \cdot C}$

If $W(\theta_s) \approx C$, then, $\tilde{W}(\theta_s) = \dfrac{1}{S}$

then, $S_{eff} = \dfrac{1}{\sum_{s=1}^{S} (\frac{1}{S})^2} = \dfrac{1}{\frac{1}{S}} = S$

$$Var_f \left(\sum_{s=1}^{S} a(\theta_s) / S \right) = \frac{Var_f(a(\theta_s))}{S}$$

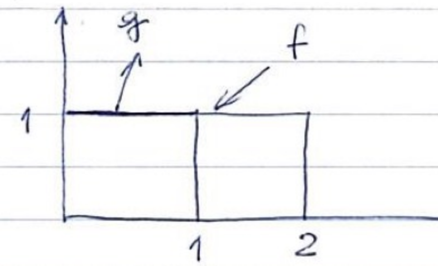
$$Var_g \left(\frac{\sum_{s=1}^{S} a(\theta_s) W(\theta_s)}{\sum_{s=1}^{S} W(\theta_s)} \right) \approx \frac{Var(a(\theta))}{S_{eff}}$$

But S_{eff} is not reliable

example3):

$f(\theta) = I(0 \le \theta \le 2)$

$g(\theta) = I(0 \le \theta \le 1)$



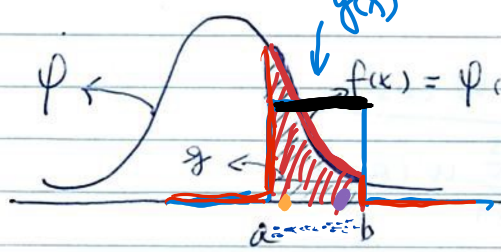
all $\theta_s \sim \text{unif}([0, 1])$

$W(\theta_s) = 1 \quad S_{eff} = S$

example 4. Estimate integral

$$\int_a^b \varphi(x)\,dx = I$$

where $\varphi = \text{dnorm}(x)$, $X \sim N(0,1)$



$f(x) = \varphi(x) \cdot I(a \le x \le b)$

$$A = \int_a^b a(x) \cdot \frac{\varphi(x)}{I}\,dx$$

$$= E_f(a(x))$$

$f(x) = \varphi(x) \cdot I(a \le x \le b)$

$$c_f = \int_{-\infty}^{+\infty} f(x)\,dx = P(a < X < b)$$

Option 1:

$g(x) = I(a \le x \le b)$, $C_g = b-a$

Suppose we have $X_s \sim g$ (unif $([a, b])$)

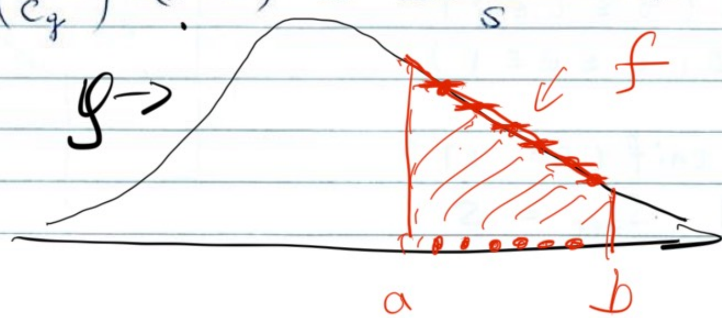
$$W(X_s) = \frac{f(X_s)}{g(X_s)} = \varphi(X_s)$$

$$\widehat{\left(\frac{C_f}{C_g}\right)} = \frac{\sum_{s=1}^{S} W(X_s)}{S} = \frac{\sum_{s=1}^{S} \varphi(X_s)}{S}$$

$$\hat{A} = \frac{\sum_{s=1}^{S} a(X_s)\varphi(X_s)}{\sum_{s=1}^{S} \varphi(X_s)}$$

we know $C_g = b - a$

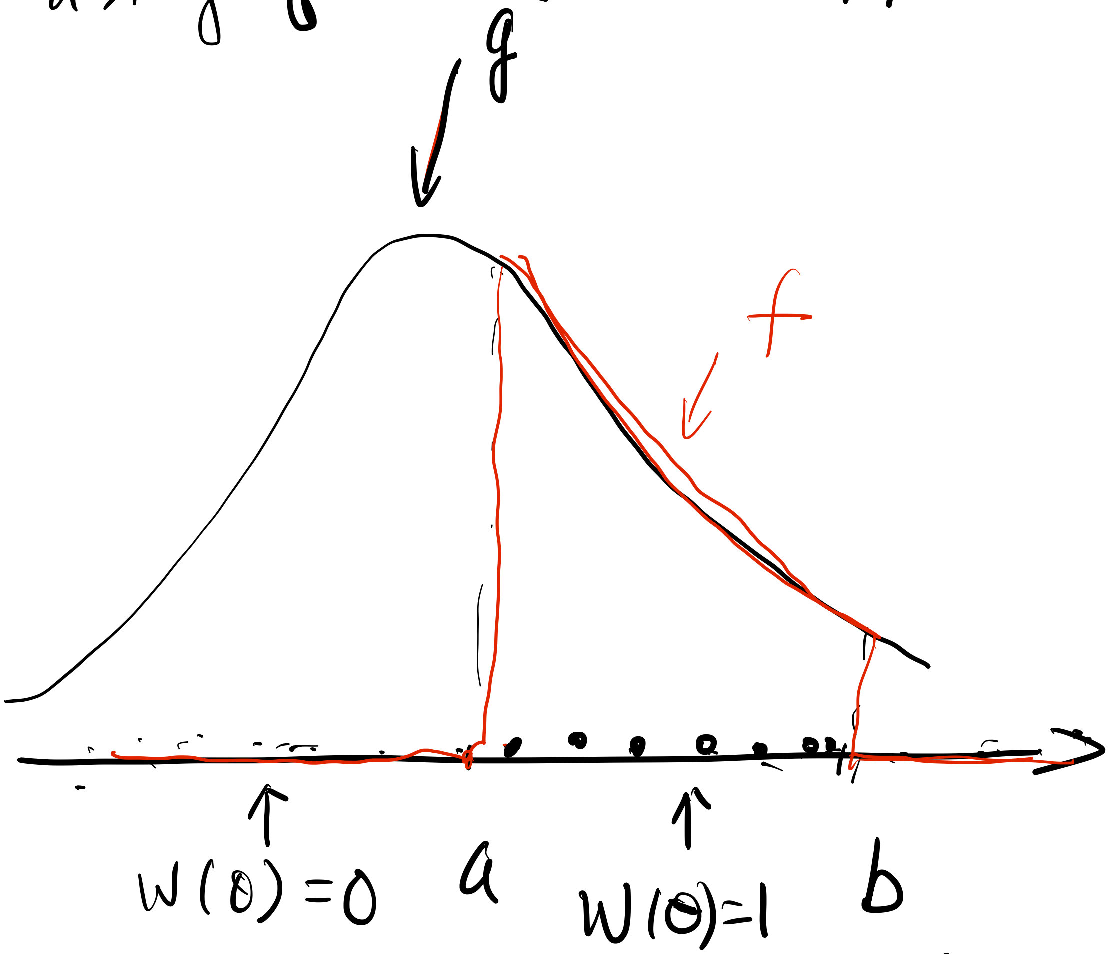
$$\hat{C}_f = \left(\frac{\hat{C}_f}{C_g}\right) \cdot (b-a) = \frac{\sum_{s=1}^{S} \varphi(X_s)}{S} \cdot (b-a) = \sum_{s} \varphi(X_s) \cdot \frac{b-a}{S}$$



randomized midpoint rule

Option 2:

using $g(x) = \varphi(x)$ to approx. f



$\theta_1, \dots, \theta_n \sim g$, $c_g = 1$

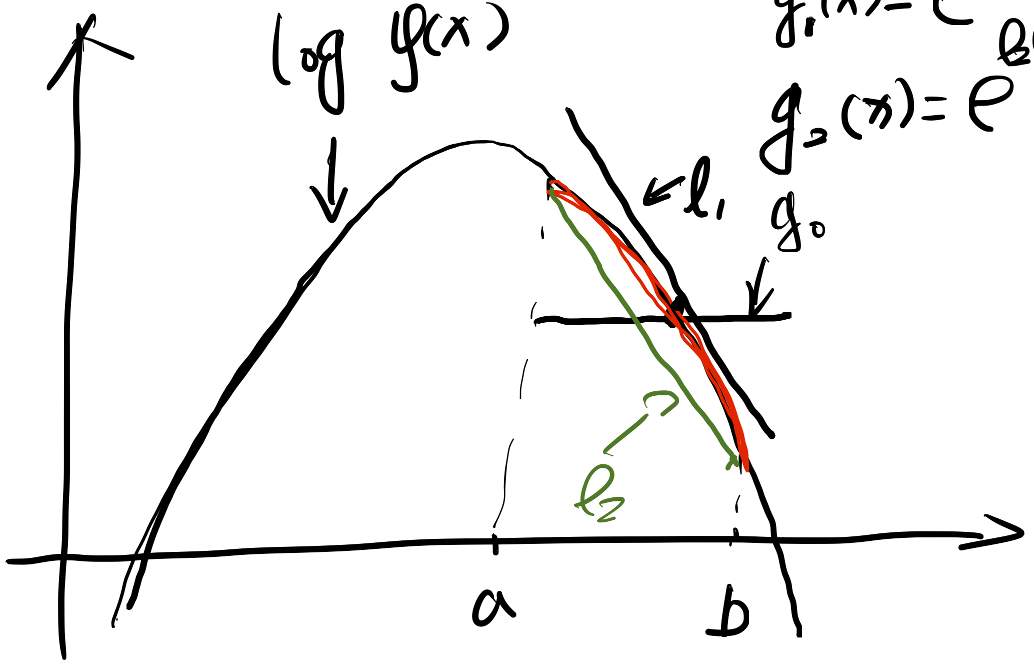
$$\widehat{\left(\frac{c_f}{c_g}\right)} = \frac{\sum w(\theta_i)}{n}$$

$$= \frac{\#(a < \theta_i < b)}{n}$$

More choices of g(x)

$g_1(x) = e^{l_1(x)}$

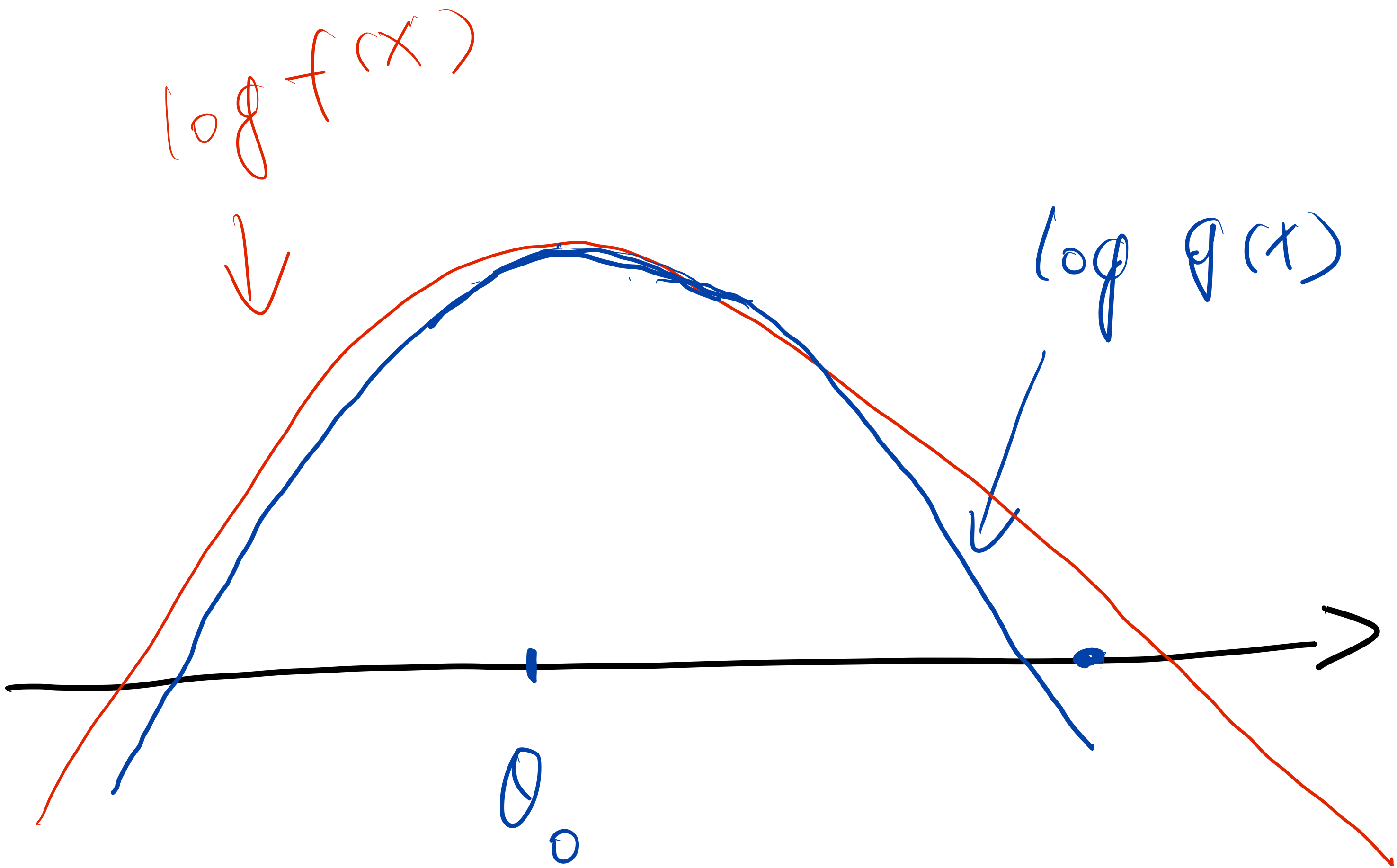
$g_2(x) = e^{l_2(x)}$



$$F(x) = \frac{\int_a^x e^{l_1(t)}\,dt}{\int_a^b e^{l_1(t)}\,dt}$$

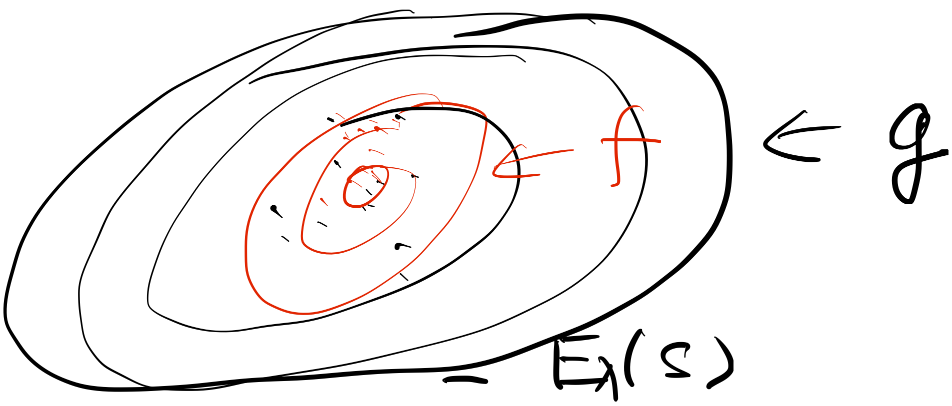
Inverting-CDF

Example: using quadratic functions to approximate $\log f(x)$.



Advanced Examples

- Free energy



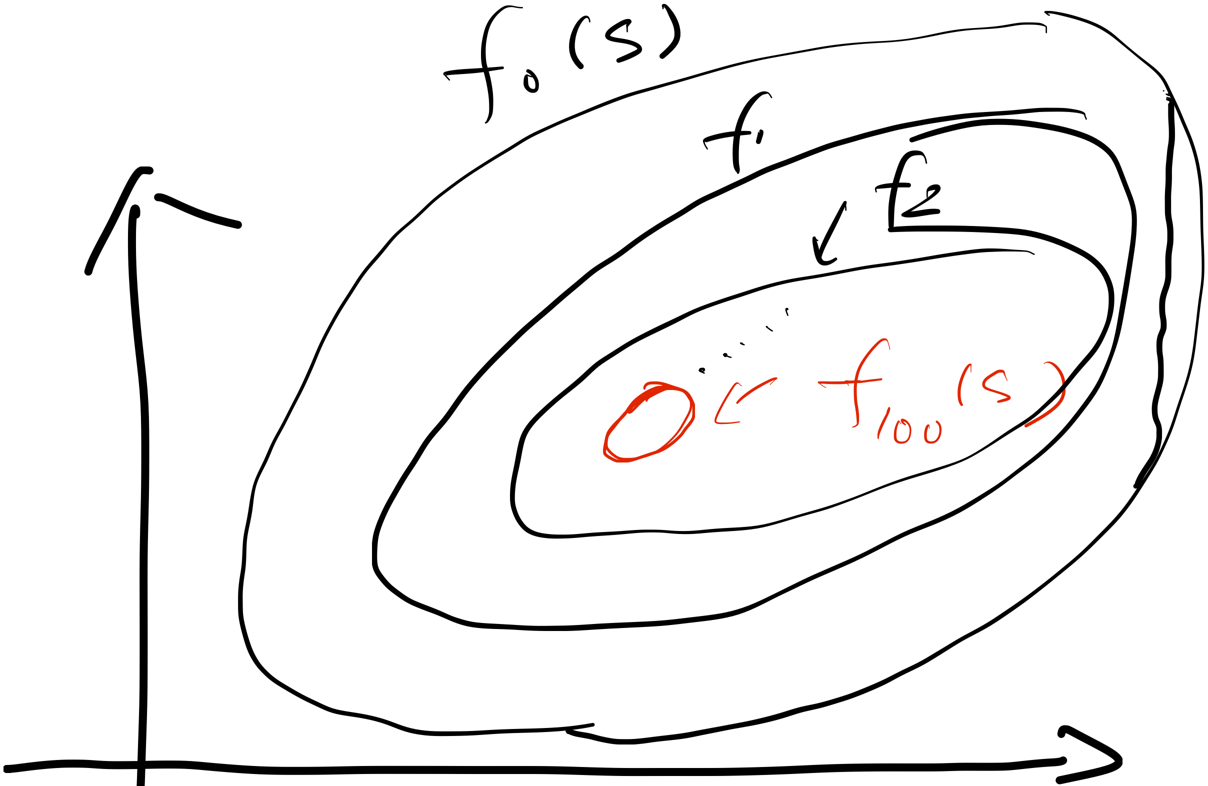
$$f_\lambda(s) = e^{-E_\lambda(s)}$$

$$F(\lambda) = \int f_\lambda(s)\, ds$$

$$f_0(s) = g(s) = e^{-E_0(s)}$$

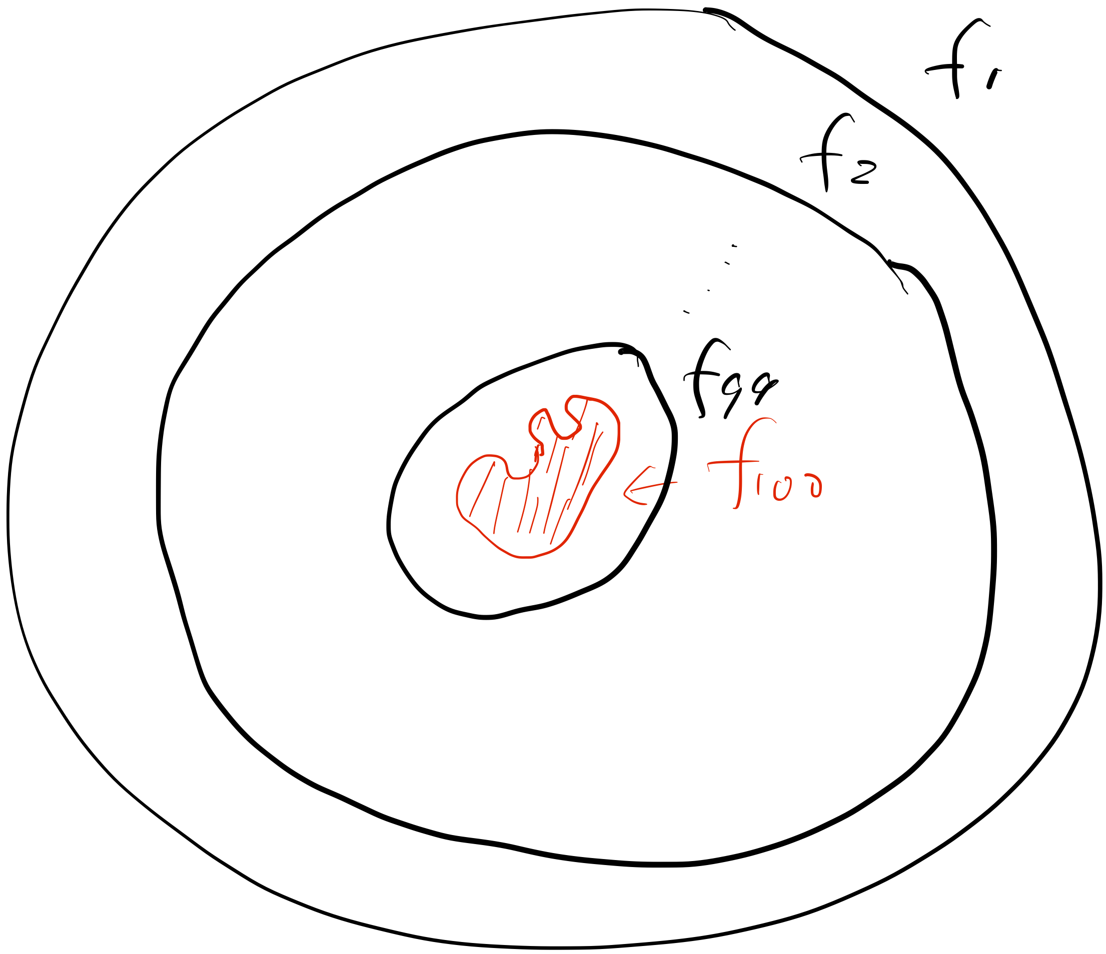
$$\widehat{\frac{F(\lambda)}{F(0)}} = \sum_s \frac{f_\lambda(s)}{f_0(s)} \Big/ n$$

with $s \sim f_0(s) = g(s)$



$$\frac{\hat{\bar{F}}_{100}}{\bar{F}_0} = \frac{\hat{\bar{F}}_{100}}{\bar{F}_{99}} \cdot \frac{\hat{\bar{F}}_{99}}{\bar{F}_{98}} \cdots \frac{\hat{\bar{F}}_{1}}{\bar{F}_{0}}$$

A Toy Example: Estimating the Area of the Red Area



Approximating Cross-validation

$y_1, \ldots \boxed{y_i} \ldots y_n$

$$f(\theta \mid y_1, \ldots, y_n)$$

$$\propto g(\theta) = \prod_{i=1}^{n} p(y_i \mid \theta) \cdot \pi(\theta)$$

Full data posterior

$$f_{-i}(\theta \mid y_1, \ldots, y_{i-1}, y_i, \ldots y_n)$$

$$\propto f(\theta) = \prod_{j \neq i} p(y_j \mid \theta) \cdot \pi(\theta)$$

$$w(\theta) = \frac{f(\theta)}{g(\theta)} = \frac{1}{p(y_i \mid \theta)}$$